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Просто закопай свой мусор

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Translate

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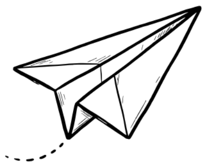
Что, если все наши представления об утилизации отходов ошибочны?

Каждого ребенка учат, что свалка — худшее место, куда может попасть наш мусор. Это печальное последствие неумения сортировать отходы: гора мусора, из которой просачиваются химикаты и неприятные запахи, навсегда отравляющие землю.

Что, если эта картина неверна? Что, если правильно организованная свалка на самом деле является одним из самых чистых, дешевых и экологичных способов утилизации отходов? Оказывается, для большей части того, что мы выбрасываем, скромная свалка предпочтительнее контейнера для вторсырья.



Одноразовые вещи часто лучше многоразовых

Аргументы против захоронения отходов делятся на две части. Первая заключается в том, что существуют эффективные, дешевые или экологичные способы практического повторного использования или переработки материалов, а не их захоронения, то есть разумная альтернатива существует. Вторая заключается в том, что захоронение отходов само по себе неустойчиво или вредно для окружающей среды, а значит, нам нужна альтернатива. Оба утверждения ложны.

Вполне естественно полагать, что многоразовые вещи всегда лучше. Но эта интуиция рассматривает разные материалы как взаимозаменяемые и игнорирует затраты на их производство. Если учесть затраты на энергию, рабочую силу и другие ресурсы, то повторное использование окупается лишь в отдельных случаях.

На производство крошечных кусочков пластика почти не затрачивается энергия. На производство полиэтиленовой смолы для одного тонкого пластикового пакета, а затем на его раздувание и запайку уходит около 0,6 мегаджоуля, примерно столько же энергии, сколько требуется для кипячения чайника. В отличие от пластика, производство хлопка требует огромных затрат энергии, отчасти потому, что хлопковые мешки должны быть намного толще, чтобы выполнять свою функцию, и весить около 200 граммов по сравнению с восемью граммами полиэтилена. Вместо простого процесса выдува и запайки хлопок нужно превратить в пряжу, соткать из нее ткань, подготовить, высушить, отбелить, покрасить и обработать, что требует около 17,5 мегаджоулей, то есть примерно в 29 раз больше энергии.

В общей сложности, по подсчетам Агентства по охране окружающей среды Великобритании, обычную хлопковую сумку нужно использовать повторно 173 раза, чтобы она стала экологичнее пластика. Аналогичный орган в Дании получил аналогичные результаты. Достаточно изношенные хлопковые сумки могут быть экологичнее пластика, но только для тех, кто использует их постоянно.

Та же закономерность прослеживается почти везде, где мы пытались заменить одноразовые товары долговечными. Сталь более энергозатратна, чем пластик, и для изготовления стальной трубочки требуется 10 граммов стали, в то время как для пластиковой трубочки — всего 0,2 грамма пластика. Таким образом, даже если стальную трубочку никогда не мыть (горячая вода также требует больших затрат энергии), для того чтобы стальная трубочка потребовала меньше ресурсов, чем пластиковая, ее нужно использовать 150 раз. А кто будет ее мыть?

A ceramic mug only beats a polystyrene cup on energy use after 1,000 uses, and even then only if your dishwasher is fairly efficient. Run a less efficient one and the disposable cup wins on energy forever. Cloth diapers feel greener than disposables, but once you account for the cotton, the laundering, the hot water, and the detergent, the UK Environment Agency found the two are roughly tied in overall demand on resources. A returnable milk bottle takes much more energy to produce and much more fuel to transport than a plastic carton, and has to make around ten or more trips to use less energy than a bottle made from polyethylene plastic.

The case for recycling to avoid using virgin material is much stronger when it comes to metals. Recycling 1 kilogram of steel saves 1.4 kilograms of iron ore and uses 65 percent less energy. For aluminum, the energy savings from recycling can be as high as 95 percent. This is because making metal from scratch requires extremely energy-intensive chemical reduction of ores, while recycling primarily involves remelting metals already in their elemental form. Because recycling metals is cheaper than mining, refining, and smelting new ones, the world does an awful lot of it. About 30 percent of the metal used in steel production globally is now recycled scrap.

Are we running out of materials?

The best case for recycling is where the materials themselves are scarce, and we can more cheaply reuse them than finding more of them. Batteries and electronics contain metals like copper, cobalt, lithium, and nickel. While we are not about to run out of any of these materials globally in a strict sense, their supplies are limited in practical terms. Plenty of copper ore remains in the ground, but it is getting harder and more expensive to extract as the highest quality and most accessible deposits have been depleted. In the case of cobalt, lithium, and nickel, production is geographically concentrated, meaning countries without natural supplies of their own have an interest in recycling in order to limit their dependence on imports. All of these, especially copper and nickel, are highly recyclable, with little loss of quality over multiple reuses. For these, recycling is sensible.

By contrast, there is no danger of running out of paper. We can always grow more trees, and that is in fact a decent way of removing carbon from the atmosphere. The silicon used to make glass is one of the most abundant elements on Earth.

American plastic comes from ethane, extracted from natural gas. The alternative to making plastic with it is to leave it in the gas stream, slightly increasing the heat it generates. One kilogram of ethane, left in the stream and made into electricity, could power a typical US house for a few hours, or it could be made into 100 shopping bags. The US has estimated reserves of gas that could last for nearly a century at current rates, and this could increase further with the discovery of new deposits.

The rest of the world makes plastics from propane, butane, and naphtha, meaning from crude oil. We will eventually exhaust this source, but not anytime soon. What's more, plastics use very little, which is why carrier bags are still so cheap. One 13-kilogram propane cylinder, the kind a family might use to power their gas grill, could instead be made into about 1,600 typical eight-gram plastic carrier bags, three years of a family's ten-shopping-bag weekly shops.

There ain't no such thing as a free recycle

Recycling is a straightforward process conceptually: collect waste, reprocess it, and turn it back into useful material. That picture breaks down once you look at what happens to waste after collection. Much of what is sent for recycling is never recycled. About 25 percent of items placed in American recycling bins are contaminants: for example, lids on containers, food, or damp paper. The number may be even higher, as the vast majority of US states do not consistently track contamination data. This can make the process energy or labor intensive and therefore expensive.

Reducing the contamination rate is challenging. Education programs have proven largely ineffective. The best single method for discouraging recycling contamination is cart tagging, where waste collection crews check the contents of a recycling bin and leave it unemptied if contaminated, affixing a tag explaining what was wrong. Despite being the single most forceful and labor-intensive curbside method, a cart-tagging program in Ohio succeeded in reducing contamination by only 29 percent over five rounds.

The imperfect recycling process means that putting low-value plastics (such as plastic film, black plastic food trays, or small items like bottle caps and straws) into the recycling bin can sometimes be actively harmful. Since these items are hard to recycle and are often screened out and sent to landfill, they may actually add to the contamination rate (as well as the time and labor needed for processing) without ever becoming useful recycled material.

Many mid-value plastics, like the polypropylene used in plastic bottles or butter tubs, require intensive washing. This removes contaminants but also detaches potentially recyclable plastic fragments, meaning the resulting mulch must be filtered. After the washing, the plastic is melted and filtered for contaminants a second time before being turned into pellets. Yield losses can be as high as 45 percent.

While the environmental case can hold up for things like plastic soda bottles, where minimal processing is needed, it quickly drops off as you go down the quality scale. For mid-quality plastics like butter and margarine tubs, recycling one ton of plastic prevents the emission of 0.43 tons of carbon dioxide compared with using virgin material. At the same time, recycled tub-grade plastic typically costs hundreds of dollars more per ton than virgin plastic. This means that consumers are effectively paying over a thousand dollars to save one ton of carbon dioxide. By contrast, direct air capture, which removes carbon dioxide from the atmosphere, currently costs an average of $490 per ton, and area specialists hope to lower the cost to $100 per ton in the coming decades. Recycling has its virtues, but recycling plastic to save energy just doesn't add up.

Your plastic toothbrush isn't killing turtles

Pro-recycling campaigns portray a world where unrecycled plastic bags, straws, and toothbrushes choke the world's oceans. This worry has led governments to restrict or disincentivize single-use plastics. But much of this is environmental theater.

Richer countries have near-universal refuse collection and almost none of their waste winds up in the ocean. In low-income countries, however, over 90 percent of waste is improperly managed, and typically ends up dumped, burned, or escaping into waterways. Plastic flows into the ocean overwhelmingly come from a small handful of rapidly growing Asian economies. Approximately a hundred rivers, mostly in South and South-East Asia, account for half of ocean plastic. And a substantial share of plastic in the ocean isn't domestic waste at all, but rather discarded fishing gear.

Ocean litter is a huge environmental problem, and solving it means building cheap, effective, and safe waste management in those countries where it originates (plus perhaps cleaning the oceans up). If you live in a Western country, throwing your trash into the recycling bin rather than general waste makes no difference at all to the amount of flotsam in the ocean: whichever bin you choose, that's not where it will end up.

Landfill is pretty clean

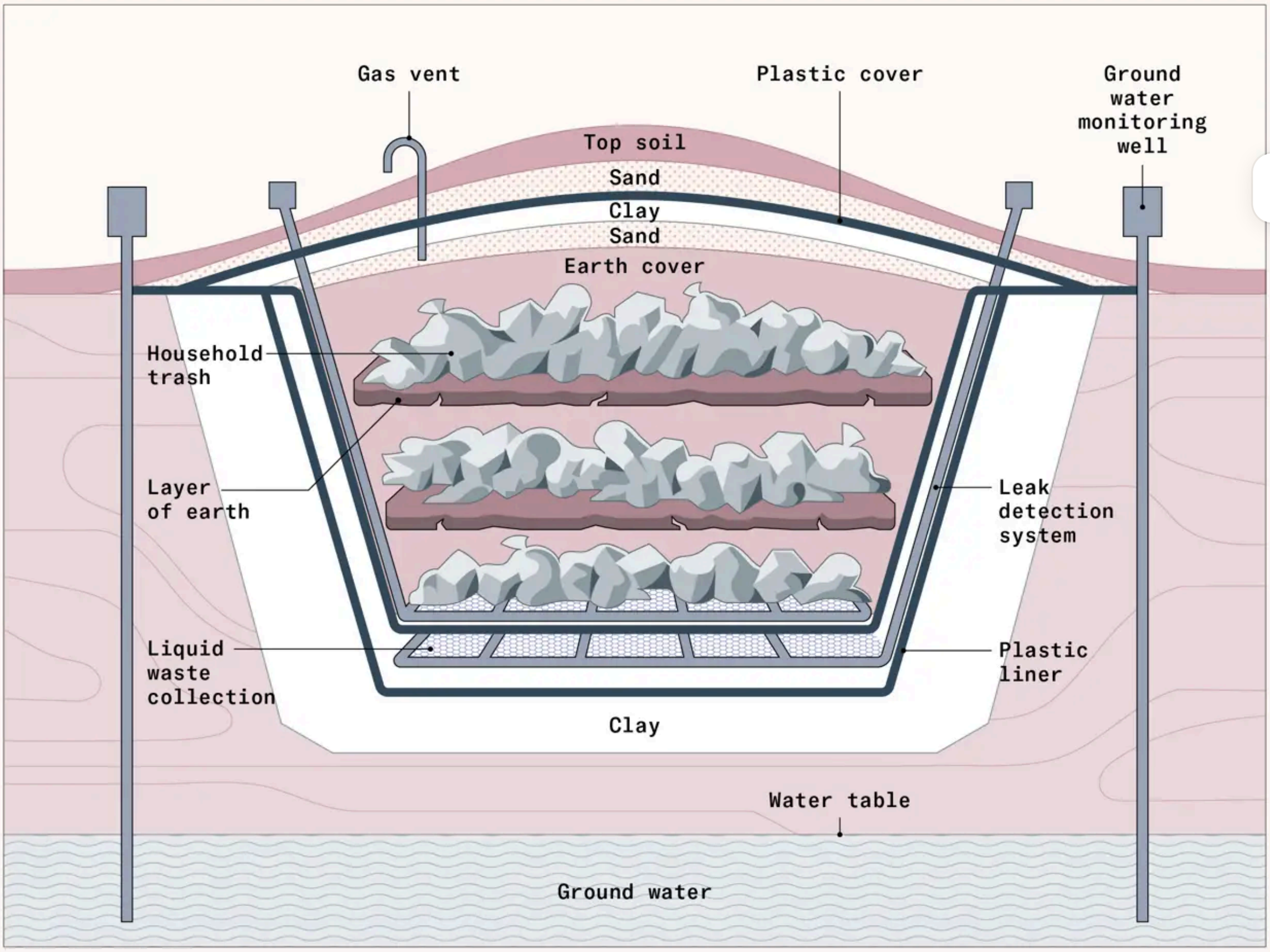
Even though recycling or reuse is energy intensive, it would be worth doing it if landfill was an environmental disaster. But most people's image of landfill is severely out of date. They envisage a rotting mound of waste, dumped in an open hole in the ground and leaching dangerous chemicals into the soil. In most of the developed world, this hasn't been true for decades. Engineers build modern landfills on thick layers of clay and tough plastic liners that contain the contaminated liquid that forms when rainwater filters through decomposing waste.

Operators place waste in thin layers, compact it tightly, and cover it regularly to control smells, litter, pests, and fire risk. As food and garden waste break down, gas wells pull methane out of the landfill body and route it to flares or generators instead of letting it escape into the air. This typically captures around 75 percent of the methane produced, and sometimes as much as 95 percent. Drainage layers and pipes collect contaminated liquid and send it for treatment, while operators monitor the surrounding ground and water, both throughout the site's life and long after it closes. Regulators shut down badly run landfills.

Much of the leakage that happens in landfills is from legacy sites with poorer liner systems. New York's environmental regulator reviewed groundwater monitoring for 31 post-1988 landfills, which had collectively operated for about 450 years, and found no adverse impact on groundwater quality.

How a modern landfill works

A modern landfill is engineered to contain waste, and has layers designed to prevent leaks, capture gas, and protect groundwater.



Adapted from the Indiana state government

Out of the garbage can, into the fire

There is another option: burning trash. Perhaps counterintuitively, this can be a very environmentally friendly option and has the advantage of reducing the waste dramatically in volume. Controlled incineration, which burns waste in a sealed, high-temperature furnace, uses systems to catch pollutants, and is more efficient than landfill at containing methane, preventing virtually any from escaping. On top of all this, the heat released can be used to generate electricity.

Unfortunately, controlled burning is more expensive than landfill. In the US, landfill is reliably 20–30 percent cheaper. While it is possible to reduce costs by selling the electricity generated, modern incinerators are among the most expensive power plants to build per megawatt, in some cases costing an order of magnitude more than a conventional natural gas plant.

This is because burning trash comes with specific complexities. The smoke from burning trash contains mercury, lead, and acid gases, requiring expensive pollution control. The acid gases produced would dissolve normal steel, so the boilers have to be lined with either specialized ceramics or a nickel-chromium alloy. This means most municipalities opt for comparatively inexpensive landfill instead.

Cheap, uncontrolled burning is bad for both the environment and human health; up to 10 percent of air pollution in Delhi comes from incinerating waste, and burning crops during the stubble-burning season contributes another 30 percentage points.

It's okay to be a landfiller

Every year, around 500 million tons of waste are recycled. US recycling rates have more than tripled since 1980, while close to 50 percent of all waste in the European Union is now recycled. Increasing this number sits at the heart of major international environmental strategies. Yet this is unnecessary and will make us poorer in the long run. Much of the time, we should bury the waste instead.

In the developed world, though there are benefits to recycling, other waste disposal options are perfectly good alternatives. Modern, well-run landfills cause minimal environmental damage. We have plenty of space to put them – less than 0.01 percent of the US is used for landfill, less than 0.001 percent of Australia is, and only about 0.02 percent of Germany is. We can even build on top of old landfills, so the land area is not spent permanently. Nor are we about to run out of most of the materials commonly recycled. And because it is less resource-intensive, landfill is far cheaper than recycling.

Instead of paying more to recycle, we could spend that money far more effectively on other environmental mitigation measures like carbon capture. The main exception is metal recycling, which should likely be the primary or even exclusive focus of municipal recycling programs. Inasmuch as some landfill sites fall short of best practices, as some surely still do, improving monitoring and imposing penalties on sloppy operators is likely to pack a much bigger environmental benefit, at lower cost, than trying to divert more materials from landfill to recycling.

And while modern landfill is ideal, worse-run landfill is still an upgrade in regions where the alternative is open waste dumping or burning. It reduces human contact with waste and localizes pollution to one managed site instead of many unmanaged ones.

There is a great humanitarian and environmental case for improving waste disposal systems in developing countries. But focusing effort on expensive recycling means getting less bang for our buck. Recycling is complex to sustai it requires the successful coordination of collection, separation, contamination control, and reliable buyers for the material. If even one part of the chain breaks down, it doesn't work. By contrast, landfill is simple and cheap. Investing in landfill, even suboptimal landfills, protects more local rivers, forests, lakes, and seas.

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The text of the article incorrectly stated that making a cotton bag required 1,725 megajoules of energy. In reality, it requires about 17.5 megajoules. The dataviz used the correct data. The article has been corrected.

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